

Agar 005

The quantized output X_q is generated according to,

$$X_q = m_i \text{ if } X_{i-1} < X \leq X_i \quad (2)$$

Where $X_i = a + i\Delta \quad (3)$

And $m_i = \frac{X_{i-1} + X_i}{2}, i = 1, 2, 3, \dots, Q. \quad (4)$

✓ EIT/EIT/11/07
Result - 43.5
Average

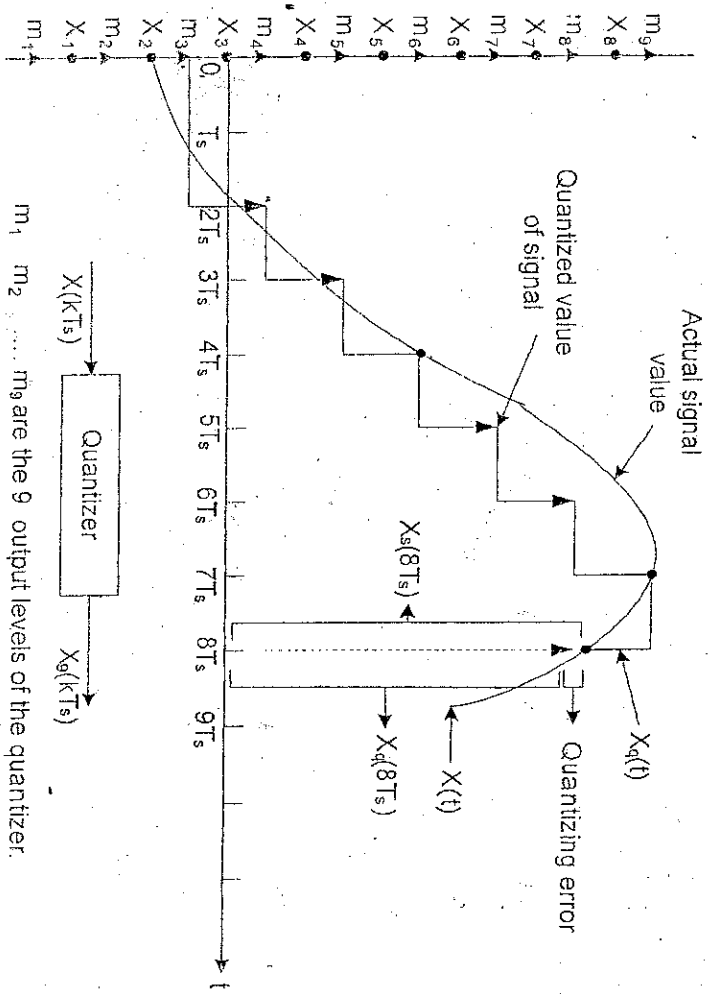


Figure 3: Input/Output Characteristics of a Linear Quantizer.

The output of the quantizer is a sequence of levels as shown in the figure as a waveform $X_q(t)$, where

$$X_q = X_q(kTs), \quad kTs \leq t < (k+1)Ts$$

Thus the quantized signal is a good approximation to the original signal. The quality of the approximation may be improved by a careful choice of x_i 's and m_i 's, such that some measure of performance may be optimized. The measure of performance that is most commonly used for evaluating the performance of a quantizing scheme is the output signal to noise power ratio (or signal to quantization error ratio).

The quantizer error (quantizing noise power) N_q for the uniform quantizer is given by,

$$N_q = E\{(X - X_q)^2\}$$

$$= \int_a^b (x - m_i)^2 f_x(x) dx$$

$$= \sum_{i=1}^Q \int_{x_{i-1}}^{x_i} (x - m_i)^2 f_x(x) dx \dots\dots\dots 1$$

Where $x_i = a + i\Delta$ and $m_i = a + i\Delta - \Delta/2$.

The signal power S_q at the output of the quantizer can be obtained from

$$S_q = E\{(X_q)^2\} = \sum_{i=1}^Q (m_i)^2 \int_{x_{i-1}}^{x_i} f_x(x) dx \dots\dots\dots 2$$

The ratio S_q/N_q gives a measure of fidelity of the linear quantizer. This ratio can be computed if the mean stationary random process of X (pdf of X) is known.

Example:

The input to a Q - step uniform quantizer has a uniform pdf over the interval $[-a, a]$. Calculate the average signal to quantizer noise power ratio at the output.

Solution: From equation (1)

$$N_q = \sum_{i=1}^Q \int_{x_{i-1}}^{x_i} (x - m_i)^2 \left(\frac{1}{2a}\right) dx$$

$$= \sum_{i=1}^Q \int_{-a+i\Delta}^{-a+(i-1)\Delta} \left(x + a - i\Delta + \frac{\Delta}{2}\right)^2 \frac{1}{2a} dx$$

$$\text{Let } y = x + a - i\Delta + \frac{\Delta}{2} \text{ then } dy = dx$$

$$\text{Hence } N_q = \sum_{i=1}^Q \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} y^2 \left(\frac{1}{2a}\right) dy = \sum_{i=1}^Q \frac{1}{2a} \left[\left(\frac{y}{3}\right)^3\right]_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}}$$

$$N_q = \sum_{i=1}^Q \left(\frac{1}{2a}\right) \left(\frac{\Delta^3}{12}\right) = \frac{Q\Delta^3}{(2a)12} = \frac{\Delta^2}{12} \quad \text{Since } Q\Delta = 2a.$$

Now the output signal power S_q can be obtained using equation (2)

$$S_q = E\{(X_q)^2\} = \sum_{i=1}^Q (m_i)^2 \frac{1}{2a} = \frac{Q^2 - 1}{12} (\Delta)^2$$

Thus the average signal to quantizer noise power ratio is,

$$\frac{S_q}{N_q} = Q^2 - 1 \approx Q^2 \quad \text{when } Q \gg 1$$

$$\left(\frac{S_q}{N_q}\right)_{dB} = 20 \log Q$$

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NON-LINEAR QUANTIZATION

This is a method of quantization in which a range of continuous random variable X is divided into Q — intervals of unequal lengths (variable step sizes).

It has two important advantages over the linear quantization method.

(i) Has a higher average signal to quantizing error ratio (or signal to quantizing noise power ratio) than the linear quantizer when the signal pdf is non — linear, which is the case in many practical situations.

(ii) The rms value of the quantizing error of a non linear quantizer is substantially proportional to the (instantaneous) sampled value X and hence the effect of quantizer noise is masked (see diagram below).

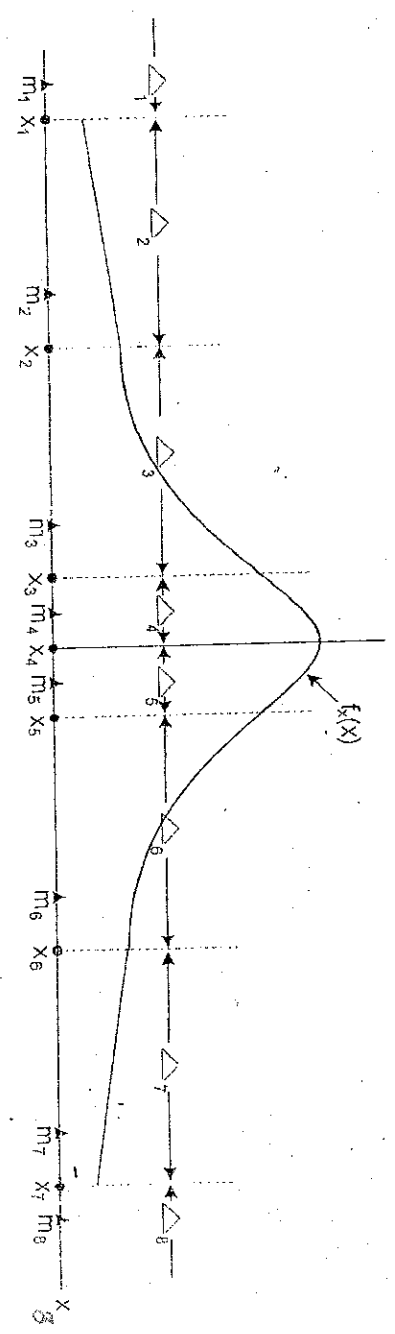


Figure 4: A non — linear quantizer for a Gaussian random variable.

$X_0 = -\infty, X_Q = \infty, Q = 8$, and $\Delta_i = \Delta_{Q+1-i}$ ($i = 1, 2, 3, 4$)

The input to the quantizer is a Gaussian random variable and the quantizer output is determined according to,

$$X_q = m_i \text{ if } X_{i-1} < X \leq X_i, i = 1, 2, \dots, Q$$

$$X_0 = -\infty, X_Q = \infty$$

The step size $\Delta_i = X_i - X_{i-1}$ is variable. The quantizer end points X_i and the output levels m_i are chosen to maximize the average signal to quantizing error ratio. In practice a non uniform (linear) quantizer is realized by sample compression followed by a uniform quantizer. Compression transforms the input variable X to another variable Y using a non — linear transformation;

$$Y = g(X),$$

Such that $f_Y(Y)$ has a uniform pdf. Then Y is uniformly quantized and transmitted. At the receiver, a complementary expander with transfer characteristic g^{-1} restores the quantized values of X . The compressor and expander taken together constitute a Compander. The most commonly used compander uses a logarithmic compression.

$\sigma = \log X$, where the levels are crowded near the origin and spaced farther apart near the peak values of X .

DIFFERENTIAL QUANTIZING SCHEME.

It has already been noted that continuous random process can be adequately represented by a sequence of its sampled values $X(kTs)$ and that the individual samples $X(kTs)$ can be approximated by a set of quantized levels. In the quantizing techniques considered thus far, each sample in the sequence $X(kTs)$ was quantized independently of the value of the preceding sample. But in many practical situations, due to the statistical nature of the message signal $X(t)$ and due to over sampling, the sequence $X(kTs)$ will consist of samples that are correlated with each other. Differential quantizing schemes take into consideration the sample to sample correlation in the quantizing process. For a given number of levels per sample, differential quantizing schemes yield a lower value of quantizing noise power than direct quantizing schemes.

Example: The message signal $X(t)$ in a communication system is a zero mean stationary Gaussian random process, and it is sampled at a rate of 10,000 samples per second ($T_s = 0.1$ m-sec). The normalized auto correlation function $R_{xx}(\tau)/R_{xx}(0)$ has a value 0.8 when $\tau = 0.1$ m-sec. Two quantizing schemes being considered are;

- A non-uniform quantizer with $Q = 32$ operating on each sample independently,
- A differential quantizer with $Q = 32$ which operates on successive differences $X(kTs) - X((k-1)Ts)$. Assuming that the mean squared error due to quantizing a normal random variable with variance σ^2 is $2\sigma^2 Q^{-2}$, find the mean squared error of the two quantizing schemes given above.

Solution: (a) For the quantizer operating independently on each sample, the mean squared error is given by,

$$N_q = 2\sigma_x^2 Q^{-2} \approx (2)(10^{-3}) \sigma_x^2$$

(b) In the differential quantizing scheme, the variable being quantized is

$$Y = X(kTs) - X((k-1)Ts) \text{ and the variance of } Y \text{ is given by,}$$

$$\sigma_y^2 = \sigma_x^2(kTs) + \sigma_x^2((k-1)Ts) - 2E[X(kTs)X((k-1)Ts)]$$

$$= 2\sigma_x^2 \left(1 - \frac{R_{xx}(T_s)}{R_{xx}(0)}\right) = 0.4\sigma_x^2$$

Hence the mean squared error of the differential quantizing is given by

$$N_q = 2(0.4) \sigma_x^2 Q^{-2} = 0.8\sigma_x^2(Q^{-2}) \approx 0.8(10^{-3}) \sigma_x^2$$

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(This is less than the error associated with direct quantizing scheme)

DIFFERENTIAL PCM SYSTEMS

These systems are particularly more efficient when the sampled message signal has high sample to sample correlation. For example in transmission of picture (video) information, appreciable portions of the signal describe background information containing very little tonal variations. In such situations, when pcm is used, the code-words describe the value of the average background level. If these tonal values do not change appreciably, then essentially repeated sample values are being transmitted. One way to improve the situation is to send only the digitally encoded differences between successive samples. Thus a picture that has been quantized to 256 levels (eight bits) may be transmitted with comparable fidelity using 4 – bit differential encoding. This reduces the transmission bandwidth by a factor of 2. PCM systems using differential quantizing schemes are known as differential pcm (DPCM) systems.

DELTA MODULATION SCHEME

A differential pcm system that is particularly simple to implement results when the difference signal is quantized into two levels. The output of the quantizer is represented by a single binary digit, which indicates the sign of the sample to sample difference. This pcm system is known as delta modulation (DM)

Delta modulation systems have the merit that the electronic circuitry required for modulation at the transmitter and demodulation at the receiver is substantially simpler than the corresponding hardware needed for PCM and DPCM systems.

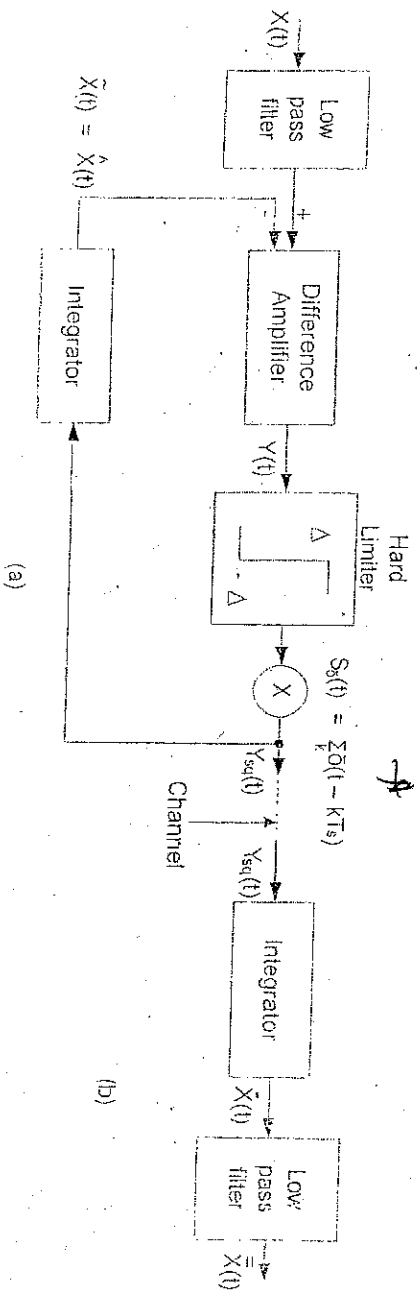


Figure 5: Block Diagramme of DM System; (a) Modulator (b) Demodulator.

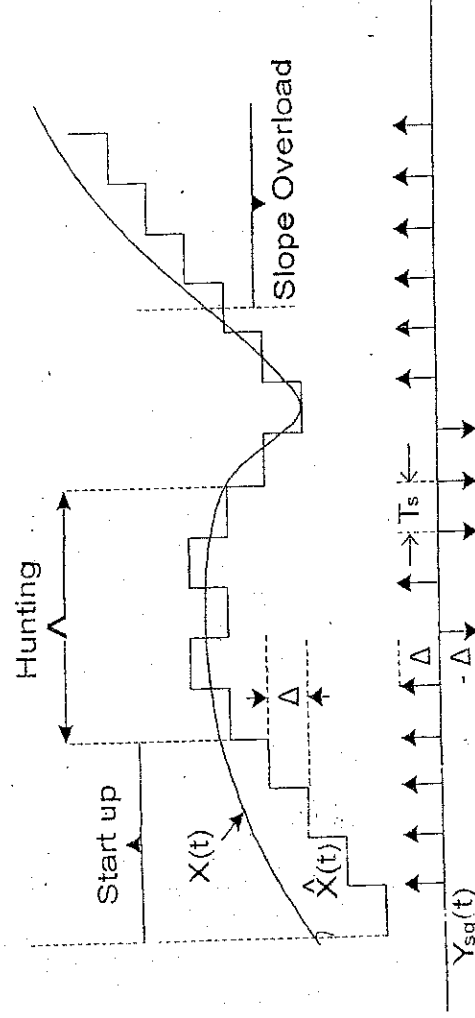


Figure 6: Delta Modulation Waveforms; $\hat{X}(t) = \bar{X}(t)$

The operation of delta modulation system is shown in the block diagram in figure 5 above, and the operation of the system may be seen by using the waveforms shown in figure 6. The message signal $X(t)$ is compared with a stepwise approximation $\hat{X}(t)$ and the difference signal $Y(t) = X(t) - \hat{X}(t)$ is quantized into two levels $\pm\Delta$ depending on the sign of the difference. The output of the quantizer is sampled to

$$\text{produce, } Y_{sq}(t) = \sum_{k=-\infty}^{\infty} \Delta \text{sgn}[X(kT_s) - \hat{X}(kT_s)]\delta(t - kT_s) \dots\dots\dots (1)$$

The stepwise approximation $\hat{X}(t)$ is generated by passing the impulse waveform in equation (1) through an integrator that responds to an impulse with a step rise. Since there are only two possible impulse weights in $Y_{sq}(t)$, this signal can be transmitted using a binary waveform. The demodulator consists of an integrator and a low-pass filter.

In practical delta modulation system, the low-pass filter in the receiver will by itself, provide an approximate measure of integration; hence the receiver integrator can be eliminated to allow the filter to do the integration. At the transmitter, the sampling waveform $S_d(t)$ need not be an impulse waveform but a pulse waveform with a pulse duration that is short in comparison with the interval between pulses. Also the transmitter integrator need not be an ideal integrator. A simple RC low-pass filter will suffice. These simplifications reduce the complexity of the hardware in Delta Modulation systems considerably.

~~DIFFICULTIES ASSOCIATED WITH DM SCHEMES~~

Problems that occur with the use of delta modulation scheme to transmit analog signal can be deduced from the waveform shown in figure 6. Suppose at the starting period, $\hat{X}(i) < X(i)$ so that the first impulse has a weight of Δ . When this impulse is fed through the integrator, it produces a step change in $\hat{X}(i)$ of height Δ . This process continues until $\hat{X}(i)$ exceeds $X(i)$.

Hunting

During the start up interval, the receiver output will differ considerably from the message signal $X(i)$ until $\hat{X}(i)$ approach $X(i)$ and equalize. After the start up period (transient period) $\hat{X}(i)$ exhibit a hunting behavior when $X(i)$ does not vary (remains constant). Hunting leads to idling noise. The sampling rate in a delta modulation scheme will normally be much higher than the Nyquist rate and hence the rectangular idling noise waveform can be filtered or smoothed out by the receiver filter.

Slope Overloading

It increases more than what it supposed to be

Another problem associated with the DM scheme is slope overloading. This overloading appears when the modulating signal changes between samplings by an amount greater than the step size Δ . Hence this type of overload is not determined by the amplitude of the modulating signal but rather by its slope – hence the name slope overloading.

PCM MULTIPLEXING (TDM/FDM)

TDM: Time Division Multiplexing, is a technique used for transmitting several analog message signals over a communication channel by dividing the time frame into slots and allocating one slot for each message signal.

FDM: Frequency Division Multiplexing is also a technique for transmitting several analog message signals over a communication channel by dividing the available bandwidth into slots, and allocating one slots for each message signal.

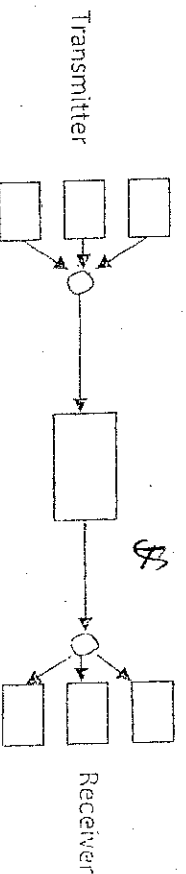
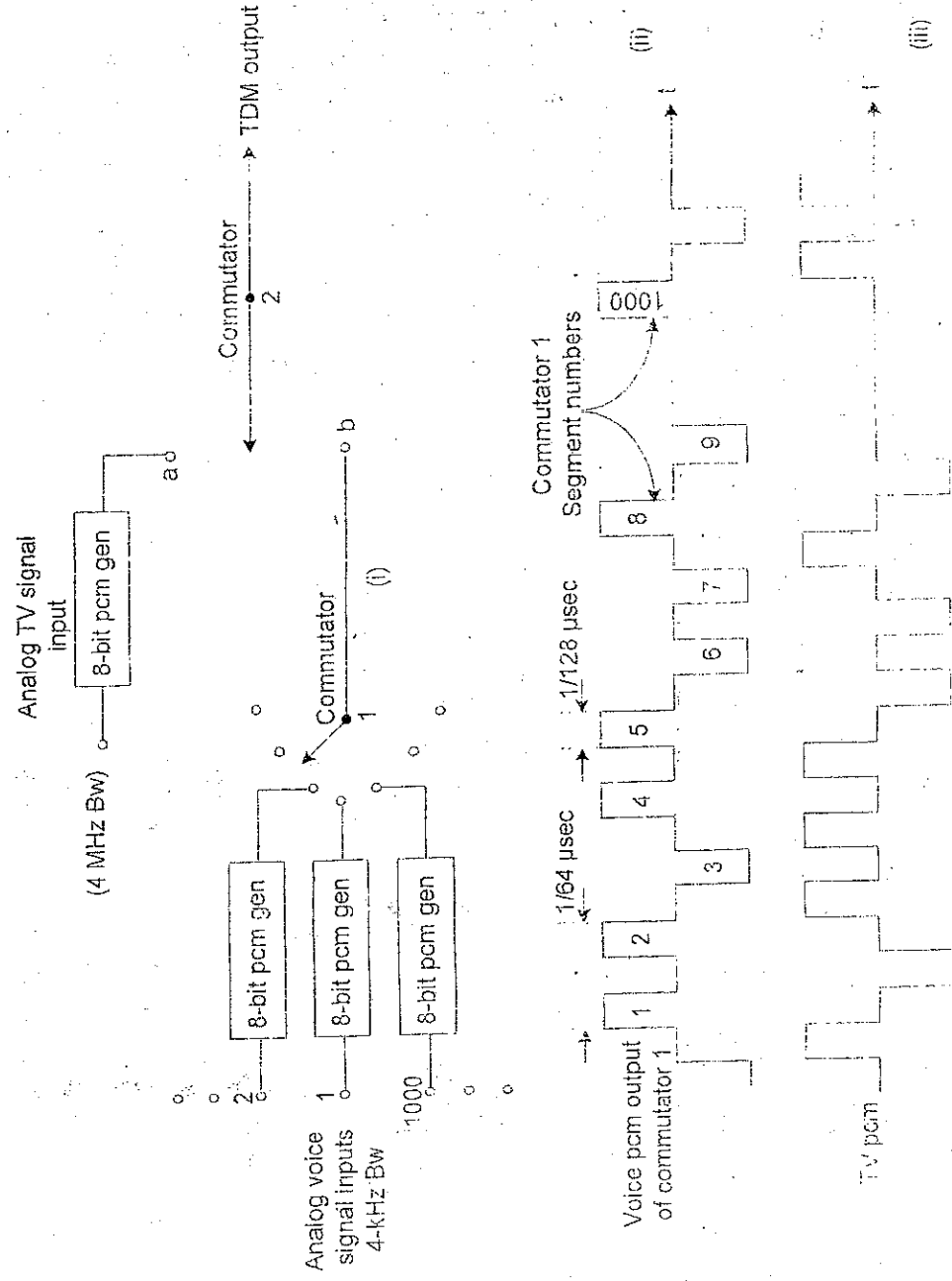


Figure 7: Block Diagramme of a 3-Channel TDM System

The diagram illustrates the principle of TDM. The input signals, all band-limited to f_x by the input filters, are sequentially sampled at the transmitter by a rotary switch or commutator. The switch makes f_s revolutions per second and extracts one sample from each input during each revolution.

At the receiver, the samples from individual channels are separated and distributed by another rotary switch called a distributor or de-commutator. The samples from each channel are filtered to reproduce the original message signal. The rotary switches at the transmitter and receiver are carefully synchronized.

Example: To illustrate multiplexing with pcm, consider a case in which 1000 voice message signals and a single television signal are multiplexed. In this problem an 8-bit pcm is used for good fidelity, since 8-bits corresponds to $2^8 = 256$ quantization levels.



Assignment

Figure 8: TDM of unequal bandwidth

Design 8 bit multiplexer that will be required to be transmitter

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A bandwidth of 4kHz is used for the voice message signals while for the TV signal a bandwidth of 4MHz is used. The multiplexing system shown in figure 8 uses two commutator switches.

For this problem, the commutator stays on a particular contact point only for the duration of one bit. The time between breaking contact at one switch point and making contact at the next is also equal to one bit. The output waveform of commutator 1 is as shown in figure 8 (ii). The pulse marked 1 is say, the first bit of 8 which will specify a sample value in channel 1. Pulse 2 is the first bit of 8 for a sample value in channel 2 etc.

Voice signals have a 4-kHz bandwidth, the sampling rate therefore is 8-kHz or sampling time is $1/8 \times 10^{-3}$ sec. Since there are 8-bits per sample, the commutator must make 8 revolutions in $1/8 \times 10^{-3}$ or 1 revolution in $1/64 \times 10^{-3}$ sec. Since there are 1000 voice signals, the time between bits in adjacent channels is $1/64 \times 10^{-6}$ sec ($1/64 \mu\text{-sec}$). The bit duration is $1/128 \mu\text{-sec}$ as shown in figure 8 (ii).

The Television signal with its 4-MHz bandwidth and 8-bit per sample generates a bit stream with bit separated by $\frac{1}{8(2 \times 4)} \times 10^{-6}$ sec or $\frac{1}{64} \mu\text{sec}$. The waveform of the TV bit stream is shown in figure 8 (iii). Note that all the operations must be done in synchronism. Commutator 2 must make contact with point b when commutator 1 is on a contact point. And when commutator 1 is between contacts, commutator 2 must make contact with point a. Similarly each pulse stream from each of the 1001 pcm generators must be timed so that individual bits fall in proper time slots without overlapping. Note that the product $4 \text{ kHz/channel} \times 1000 \text{ channels}$ yields 4 MHz which is the bandwidth of the television signal. Thus this relationship could allow easy interleaving of the voice and TV signals together. This is the reason why 1000 voice channels are used in the problem.

Exercises:



(1) Using block diagrams design a pcm multiplexing system to permit the combining of 3 signals, x_1 , x_2 and x_3 . The signals are band limited to 5 kHz, 10 kHz and 5 kHz respectively. Each signal is to be sampled at its Nyquist rate and quantized into 256 levels ($N = 8$). The commutator dwells on each signal so that the 8 bits are transmitted simultaneously.

- (a) Sketch the system. What is the maximum bit duration that may be used?
- (b) What is channel bandwidth required to pass the pcm signal.
- (c) What is the commutator speed in revolutions per second?

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(2) Three signals x_1 , x_2 and x_3 having bandwidths 5 kHz, 10 kHz and 20 kHz are each quantized into 256 levels. Each signal lasts for 10 minutes. If each signal is sampled at its Nyquist rate,

(a) How many samples are taken from each signal?

(b) How many bits are produced from each signal, if each signal is pcm encoded using 8 bits per sample?

CHAPTER TWO TRANSMITTERS AND TRANSMITTER TESTING

Introduction: The function of a radio transmitter is to convert a source of electric power into an intelligence carrying radio frequency wave. The transmitter output power is fed to an antenna which releases the electromagnetic energy from the confines of a transmission line to an electromagnetic wave travelling through space.

Exercise: How is the energy released from the radiating antenna?

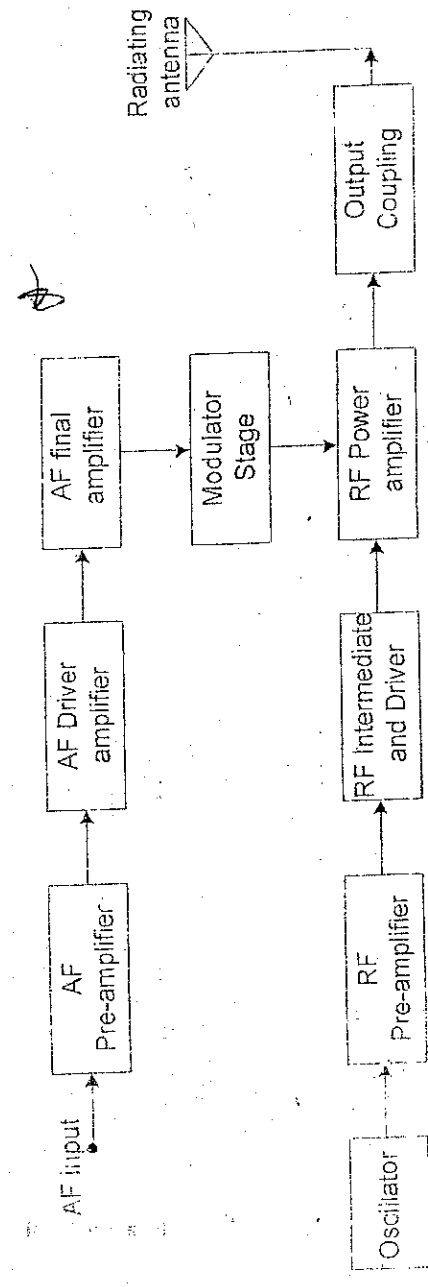


Figure 9: Block Diagram of AM Transmitter

The stages that make up the transmitter are as follows:

- (i) Oscillator Stage: Here the carrier frequency is generated. The oscillators at this stage must provide RF voltage precisely at the desired frequency, with a high order of frequency stability, free from hum or noise modulation.
- (ii) RF Amplifier: Usually the desired carrier frequency is generated at a low power level by a stable oscillator, ordinarily a crystal oscillator. This is followed by the RF amplifier that increases the power level and isolates the oscillator from the modulator.
- (iii) Modulator Stage: For the message signal to be sent out, it must be superimposed upon the carrier signal and transmitted. The process is known as modulation and the stage where this takes place is called the modulator stage.

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(iv) AF Amplifier: This set of amplifiers amplify the audio signal before modulation. This is so because the audio signal is of low electrical energy at its input. Thus at its output high energy audio signal is being delivered.

(v) RF Power Amplifier: This further increase the carrier power and the modulated signal power suitable for transmission. Tuned circuits are generally used in the input and output circuits of RF power amplifiers. They are divided into classes A, B and C according to the angle of plate current flow.

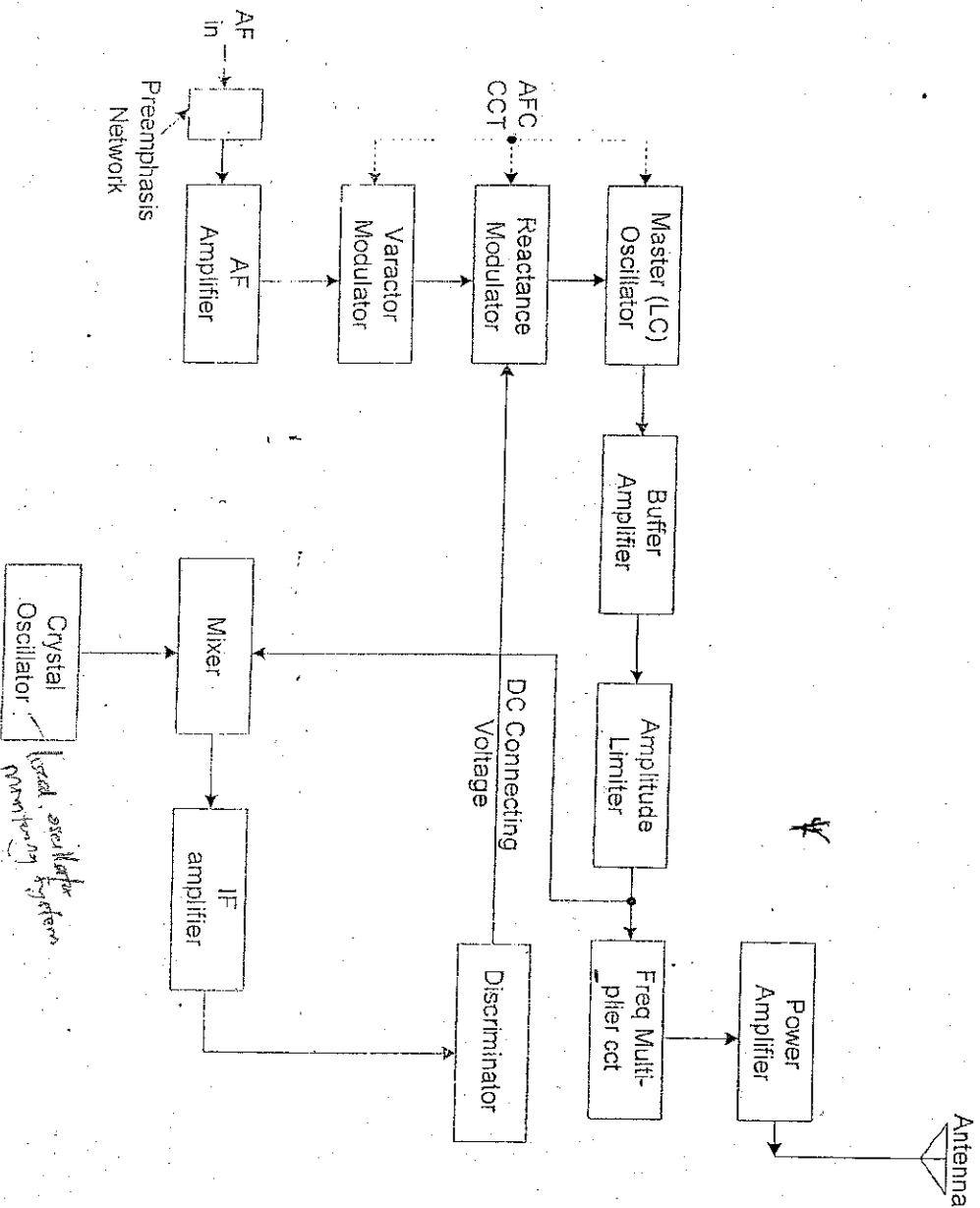


Figure 10: FM Transmitter with AFC Circuit System

Definition of FM:

This is a type of modulation in which the frequency of the carrier is made to vary in accordance with variation in the modulating signal.

Varactor Modulator: This is used to produce the frequency modulation. Most often it is employed together with the reactance modulator to provide automatic frequency correction for the FM transmitter.

Automatic Frequency Control Circuit: This is used to provide frequency stability in the system.

LC Oscillator: This constitutes the master oscillator module. It provides the required tank circuit for the reactance modulator to operate upon and also helps to boost the frequency of the modulated AF signal.

TYPES OF COUPLING FROM POWER AMPLIFIER STAGE TO THE AERIAL

Definition: A coupling network or antenna coupler is a network composed of reactances and transformers, which may be lumped or distributed. The coupling network is sent to provide impedance matching and is employed for any of the following reasons:

(i) To tune out the reactive component of the antenna impedance, to make the impedance look resistive to the transmitter, otherwise detuning will take place when the antenna is connected. This function involves the provision of variable reactances.

(ii) To provide the transmitter (and the transmission line if used) with the correct value of load resistance. This involves having one or more adjustable transformers.

(iii) To prevent the illegal radiation of spurious frequencies from the system as a whole. This function requires the presence of filtering network, generally low-pass, since the spurious frequencies are most likely to be harmonics of the transmitter's frequency.

(iv) For very low frequency transmitters whose output tank is series fed and single-tuned, the antenna coupler must prevent the dc supply from reaching the antenna. This is done to prevent antenna insulation difficulties and danger to operators which ranges from RF burns to a more lethal dc high voltage shock.

The type of coupling system depends on how the antenna is fed, whether a transmission line is used, whether it is balanced or unbalanced and the value of its standing wave ratio.

DIRECTLY FED ANTENNAS:

These antennas are coupled to their transmitters without transmission lines, generally for lack of space. To be of use, a line connecting an antenna to its transmitter ought to be at least a half-wave in length, and at least the first quarter-wave portion of it should come away at right angles to the antenna. This is difficult to accomplish especially at low frequencies, for shipboard transmitters or those on top of buildings.

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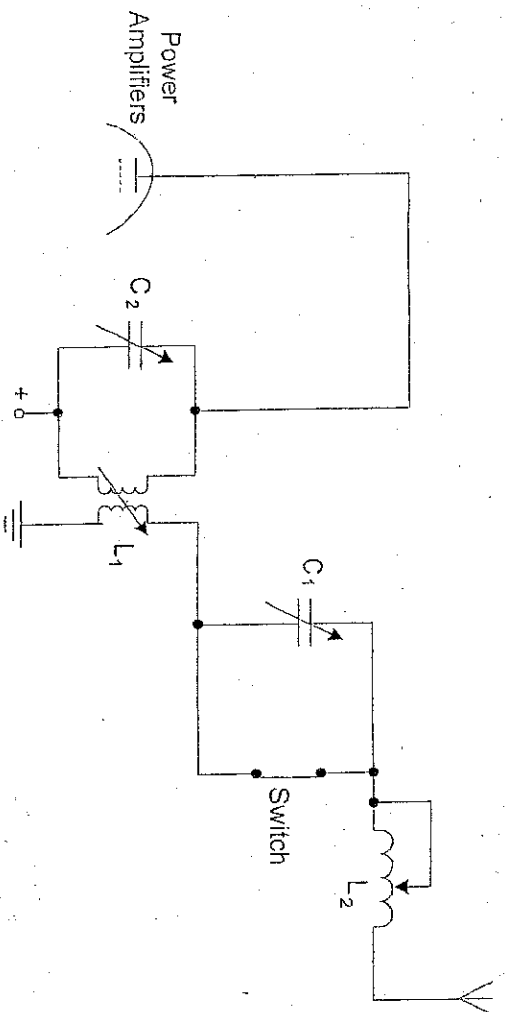


Figure 11: Direct Coupling

The figure above shows the simplest method of direct coupling. The impedance seen by the tank circuit is adjusted by moving coil L_1 or by changing the number of turns with a travelling short circuit. To tune out the antenna reactance, either C_1 or L_1 is shorted out, and the other component is adjusted to suit. This simple coupling network is not the best because it does not attenuate harmonics.

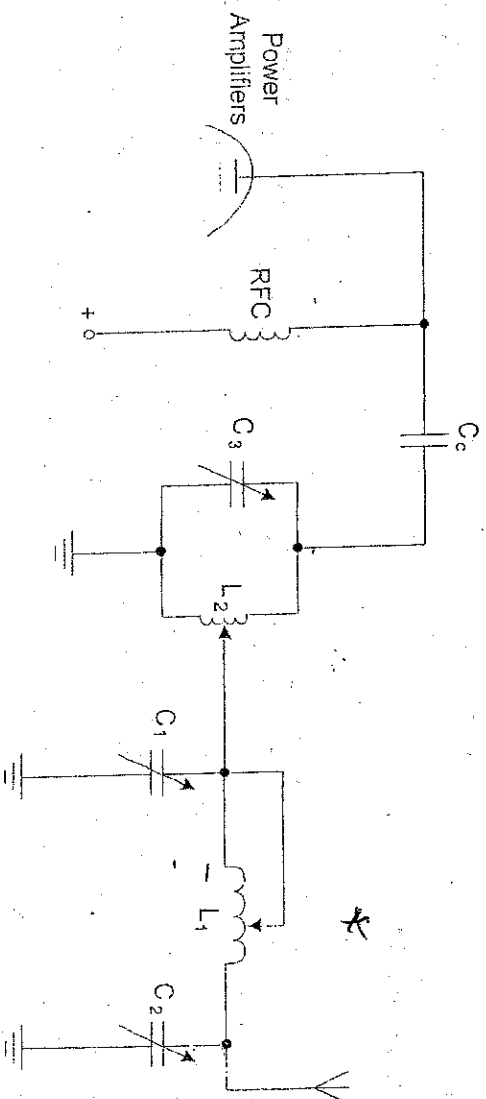


Figure 12: Direct π Coupling

Another method of direct coupling is the use of pi - coupling shown in figure 12 above. This is a much better proposition. It affords a wider reactance range and is also a low pass filter, giving adequate harmonic suppression. However, it will not provide satisfactory coupling if the antenna is very short, thus having mostly capacitive input impedance. It is better under these conditions to increase the height of the antenna.

COUPLING WITH A TRANSMISSION LINE:

The requirements are similar to the ones already considered. However, balanced lines and therefore balanced coupling networks are often used; in this case a symmetrical π – coupling network is used as shown in figure 13 below. The output tank

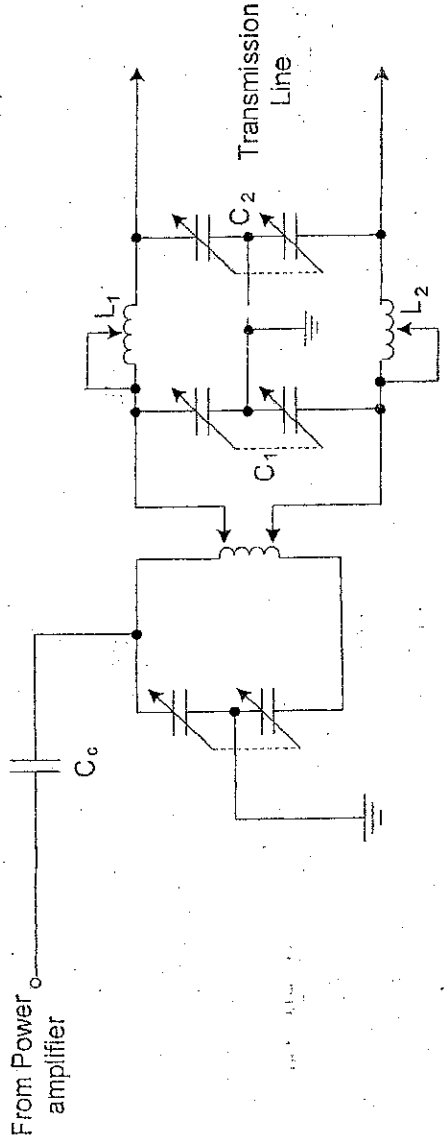


Figure 13: Symmetrical π - Coupler

is tuned accordingly and facilities must be provided to ensure that the two legs of the coupler are kept balanced. At higher frequencies a number of the lumped components can be distributed, so that quarter – wave transformers and stubs can be used.

MEASURING INSTRUMENTS FOR TRANSMITTER TESTING

There are many measuring instruments used in testing transmitters. The most general ones are the ammeters, voltmeters, power meters, frequency meters, oscilloscopes and spectrum analyzers etc.

OSCILLOSCOPES: In the history of electrical and electronic measurement, no instrument has had greater impact than the oscilloscope. Galvanometers and other mechanical apparatus capable of indicating dynamic phenomena existed long before the oscilloscope, but they were machines with slow response. However, the oscilloscope has the advantage of displaying high speed phenomena for easy observation, an asset which grew to greater importance as applications for electronics developed over the years. Today the oscilloscope is the most versatile general purpose electronic measuring instrument available for scientific investigation.

The oscilloscope is an electronic instrument used for display of electrical signals on the vertical axis compared with time on the horizontal axis. Thus the modern

laboratory oscilloscope is a time domain measurement and display instrument for analysis of electrical signals.

The function of the oscilloscope may be defined as capturing, displaying and analyzing a time-domain waveform. Traditionally, the oscilloscope has been used to capture high speed transient phenomena and display them for the engineer or technician to analyze and interpret. The great majority of scopes in use captures a waveform and faithfully portrays that waveform on the CRT for the viewer to observe. Some oscilloscopes today are capable of analyzing data and presenting digital answers for their users. Others in conjunction with computers are capable of wide varying data accumulation and analysis.

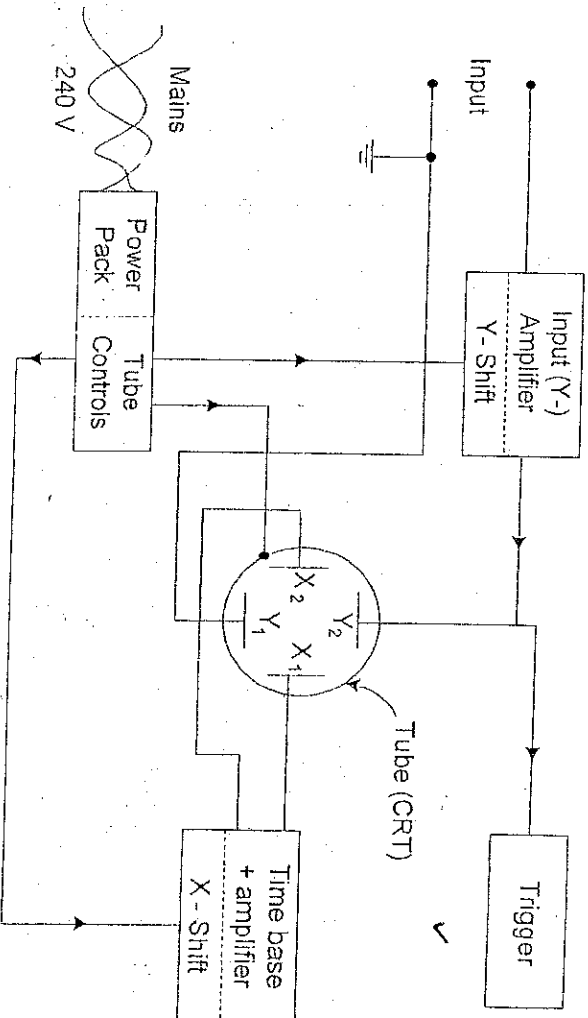


Figure 14: Oscilloscope Block Diagram

The modern time-domain oscilloscope contains the following component blocks:

- (i) Display device (CRT)
- (ii) Power pack supplies
- (iii) Tube Controls: The brightness and focus and correction of astigmatism, are controlled by varying the voltage of the appropriate tube electrode.
- (iv) Y – or input – amplifiers: This is a variable gain linear amplifier which amplifies the signal applied to the Y – plate.
- (v) Y – Shift: This is in the Y – amplifier circuit and it shifts the signal trace up or down on the screen.

- (vi) Time – base: This is a saw-tooth oscillator. In the oscillator circuit, the feedback can be adjusted with stability control until the oscillator is just not free running, and the trigger is then applied to lock the trace.
- (vii) Trigger Unit: This applies a triggering pulse to the time-base oscillator from the Y – or input voltage. The time base frequency is then synchronized (locked) with that of the input signal, so that a stationary trace is obtained on the screen.
- (viii) Time – base amplifier: This amplifies the time base voltage and applies it to the X – plates. The trace can be made to expand or contract horizontally by varying the amplification.
- (ix) X – shift: This shifts the time base horizontally to the left or right.

SPECTRUM ANALYZERS:

A spectrum analyzer visually displays (on a CRT) the amplitude of the components of a wave as a function of frequency. This can be contrasted with an oscilloscope display which shows the amplitude of the total wave (all components) versus time. Thus an oscilloscope shows the time domain while the spectrum analyzer shows the frequency domain.

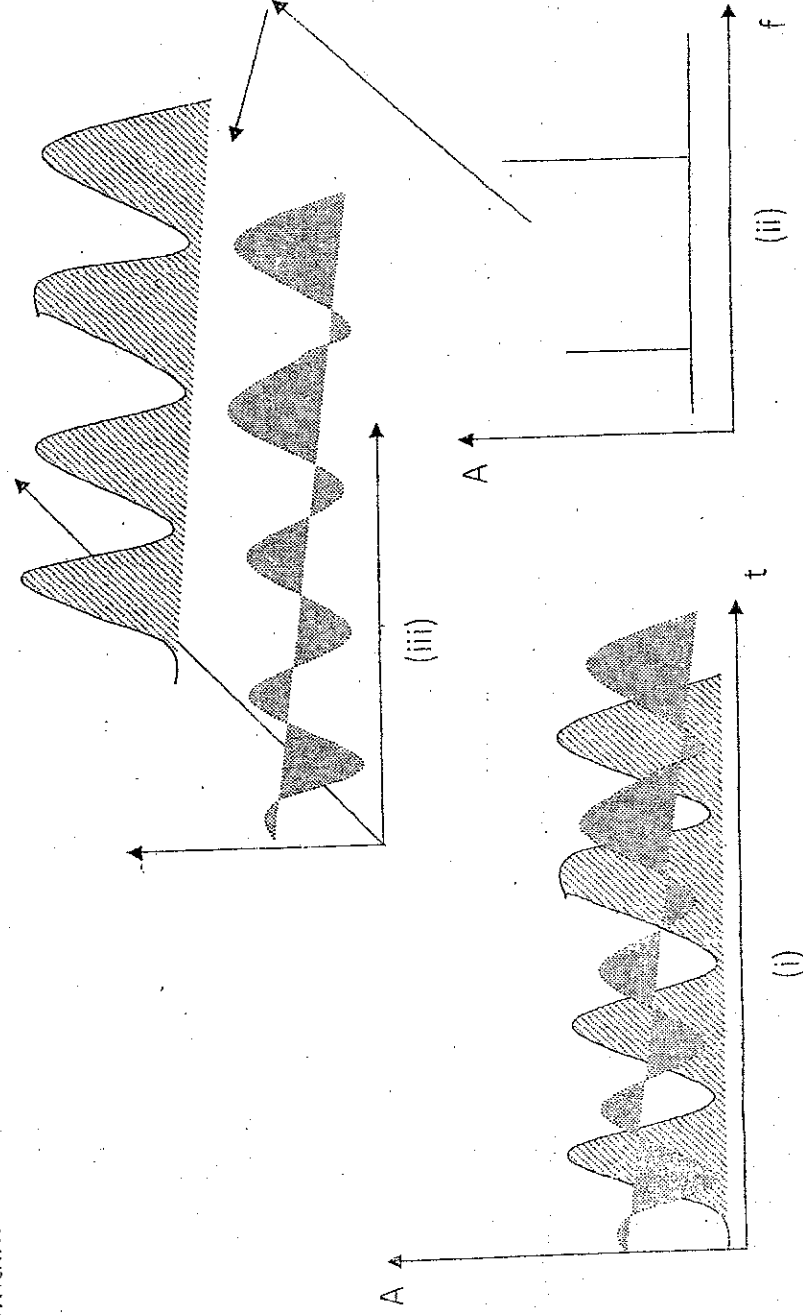


Figure 15: Spectrum Analyzer Principle

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Figure 15 above shows how a simple signal $A(t)$ looks in the time domain of an oscilloscope (i) and in the frequency-domain $A(f)$ on a spectrum analyzer (ii) and (iii). The frequency domain components are coherent that is, are exact frequency multiples and bear a finite phase relationship to each other. Also several sinusoidal components of differing frequencies that are unrelated to each other are present. These components could be signals from unrelated oscillators, which could be carrier signals of a telephone, a telemetry frequency multiplex system or perhaps several radio stations that are simultaneously present.

Time domain analysis and spectrum analysis of signals complement each other. Spectrum analyzer can measure extremely small amounts of distortion far beyond the capability of an oscilloscope by measuring very small harmonic components generated from a sinusoidal test signal by the device under test.

Spectrum analyzer applications include the measurement of signal level and the measurement of frequency and its response, harmonic and inter-modulation distortion, frequency stability, spectral purity, modulation index and attenuation.

In RF testing equipment, dummy load is used in place of actual loading conditions. The purpose is that it absorbs the excess RF voltages which are not needed by the testing equipment, thereby preventing and protecting the circuit from the damaging effects of RF voltages which could result in an unloading condition.

There is always a time delay between the transmitter and the receiver when messages are sent. This normally depends on the modulating frequency and the phase difference between the two equipment.

The Time Delay is given by,

$$T_d = \frac{d_\phi}{d_\omega}, \text{ where } \omega = 2\pi f$$

For a phase differential between the transmitter and receiver being 5.4° , the time delay will be 0.5mseconds for a modulating frequency of 30Hz. Then for a modulating frequency of 60Hz what will be the differential time delay if the phase difference between the transmitter and the receiver is 10.8° ?

Solution:

Differential time delay is given by:

$$T_d = \frac{d_\phi}{d_\omega}, \quad \omega = 2\pi f = 2\pi \times 60^\circ$$

$$d_\phi = 10.8^\circ$$

$$\therefore T_d = \frac{10.8^\circ}{360^\circ} \times \frac{2\pi}{2\pi \times 60^\circ} = 5 \times 10^{-4} \text{ sec} = 0.5 \text{ msec}$$

Exercises

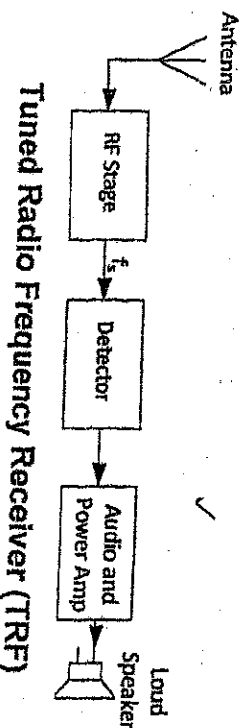
1. Calculate the differential time delay for a phase differential between the transmitter and receiver given as $270.8^\circ > 0 > 260^\circ$ for a modulating frequency of 30Hz. (Take $2\pi = 360^\circ$).
2. What is the basic circuit of a transmitter? Briefly describe the function of this circuit.
3. In the FM transmitter explain the purpose of the DC connecting voltage to the automatic frequency control (AFC) system of the transmitter. *It acts as a feedback*

RADIO RECEIVERS

Of all the radio receivers proposed at one time or the other, only two of them are of commercial significance. These are the Tuned Radio Frequency receiver and the Super-heterodyne receiver.

The Tuned Radio Frequency Receiver

It is the most elementary receiver design consisting primarily of RF amplifier, a detector, Audio and Power amplifier stages. The tuned radio frequency (TRF) receiver has three stages of RF amplification, with each stage preceded by a separate variable tuned circuit.



Sensitivity and Selectivity: Two major characteristics of any radio receiver are its sensitivity and selectivity.

Sensitivity of a radio receiver can be defined as the minimum input RF signal to the receiver capable of producing a specified audio signal at the output. The range of sensitivities for communication receivers vary from the millivolt (10^{-3} volts) region for low cost AM receivers, down to the nanovolt (10^{-9} volts) region, for ultra-sophisticated units for more exacting applications.

Selectivity of a radio receiver is defined as the extent to which a receiver is capable of differentiating between the desired signal and other frequency signals (unwanted radio signal and noise).

Problem of TRF Selectivity: A standard AM receiver can handle a signal up to a maximum of 5 kHz, which subsequently generates upper and lower sidebands extending 5 kHz above and below the carrier frequency. The total signal has a 10 kHz bandwidth. Therefore the optimum receiver selectivity is thus 10 kHz. However if a 5 kHz bandwidth were selected, the upper and lower sidebands would only extend 2.5 kHz above and below the carrier. The radio would lack full possible reception on the other hand, an excessive selectivity of 20 kHz bandwidth would result in reception of unwanted...

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$1000 = \frac{1}{2\pi\sqrt{10 \times 10^{-3} \times C}}$$

$$\frac{1000}{1} = \frac{1}{6.283 \sqrt{10 \times 10^{-3} \times C}}$$

$$6283.185 \sqrt{10 \times 10^{-3} \times C} = 1$$

$$\sqrt{10 \times 10^{-3} \times C} = \frac{1}{6283.185}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$f_r = (2\pi\sqrt{LC})^{-1}$$

$$f_r = (2\pi\sqrt{LC})^{-1/2}$$

$$\frac{f_r}{1} = \frac{1}{2\pi(\sqrt{LC})^{1/2}}$$

$$f_r \cdot 2\pi(\sqrt{LC})^{1/2} = 1$$

$$(\sqrt{LC})^{1/2} = \frac{1}{f_r \cdot 2\pi}$$

$$\frac{\sqrt{LC}}{1} = \sqrt{\frac{1}{f_r \cdot 2\pi}}$$

$$L$$

$$C = \sqrt{\frac{1}{f_r \cdot 2\pi}} \cdot \frac{1}{L} = \frac{1550}{1000}$$

$$\sqrt{10 \times 10^{-3}} = 1.59 \times 10^{-4}$$

$$(10 \times 10^{-3})^{1/2} = 1.59 \times 10^{-4}$$

$$10 \times 10^{-3} = \sqrt{1.59 \times 10^{-4}}$$

$$10 \times 10^{-3} C = 0.0126$$

$$C = \frac{0.0126}{10 \times 10^{-3}} = \underline{\underline{1.26 \mu F}}$$

$$C = \sqrt{\frac{1}{2\pi \cdot f_r}} \times \frac{1}{L}$$

$$C = \sqrt{\frac{1}{2\pi \cdot f_r L}}$$

$$C = \sqrt{\frac{1}{2\pi \cdot 1000 \times 10 \times 10^{-3}}}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}} \quad \frac{\sqrt{LC}}{1} = \sqrt{\frac{1}{f_r \cdot 2\pi}}$$

$$\frac{f_r \cdot 2\pi\sqrt{LC}}{f_r \cdot 2\pi} = \frac{1}{f_r \cdot 2\pi}$$

$$\sqrt{LC} = \frac{1}{f_r \cdot 2\pi}$$

$$(\sqrt{LC})^{1/2} = \frac{1}{f_r \cdot 2\pi}$$

$$\sqrt{LC} = \sqrt{\frac{1}{f_r \cdot 2\pi}}$$

adjacent radio signals and additional external noise that is directly proportional to the bandwidth selected. Unfortunately, TRF receivers suffer from problems of poor selectivity.

Estimating TRF Selectivity: Let us consider a standard AM broadcast band receiver that spans the frequency range from 550 to 1550 kHz. If the approximate center of this frequency range of 1000 kHz is considered, then

$$Q = \frac{f_r}{BW} = \frac{1000 \text{ kHz}}{10 \text{ kHz}} = 100$$

Note that Q is called the quality factor of the filter and it is the measure of how selective (narrow) its passband is compared to its center frequency, f_r .

Now, since the Q of the tuned circuit remains fairly constant as its capacitance is varied, a change to 1550 kHz will change the BW to 15.5 kHz.

$$Q = \frac{f_r}{BW} \quad \therefore BW = \frac{f_r}{Q} = \frac{1550 \text{ kHz}}{100} = 15.5 \text{ kHz}$$

The receiver's BW is now too large, and it will suffer from adjacent station interference and increase noise.

On the other hand, the opposite problem is encountered at the lower end of the frequency range. At 550 kHz, the BW is given by,

$$BW = \frac{f_r}{Q} = \frac{550 \text{ kHz}}{100} = 5.5 \text{ kHz}$$

The fidelity of the reception is now impaired. The maximum intelligence frequency possible is 5.5 kHz/2 or 2.75 kHz.

Exercise

✓ A TRF receiver is to be designed with a single tuned circuit using a 10 μH inductor.

- Calculate the capacitance range of the variable capacitor required to tune from 550 to 1550 kHz.
- The ideal 10 kHz BW is to occur at 1100 kHz. Determine the required Q .
- Calculate the BW of this receiver at 550 kHz and 1550 kHz.

Hint use the formula to calculate C

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$\overline{x} = 40.6$$

$$\overline{x} = 45.5$$

$$\overline{x} = 45.5$$

AM Detection: If the AM signal is passed through a nonlinear device, difference frequencies between the carrier and sidebands will be generated and these frequencies are, in fact the intelligence.

Detection of amplitude modulated signals requires a nonlinear electrical network. An ideal nonlinear curve for this is one that affects the positive half-cycle of the modulated wave differently than the negative half-cycles. This distorts an applied voltage wave of zero average value so that the average resultant current varies with the intelligence signal amplitude. The curve shown in Figure 19(a) is called an *ideal curve* because it is linear on each side of the operating point P and does not introduce harmonic frequencies.

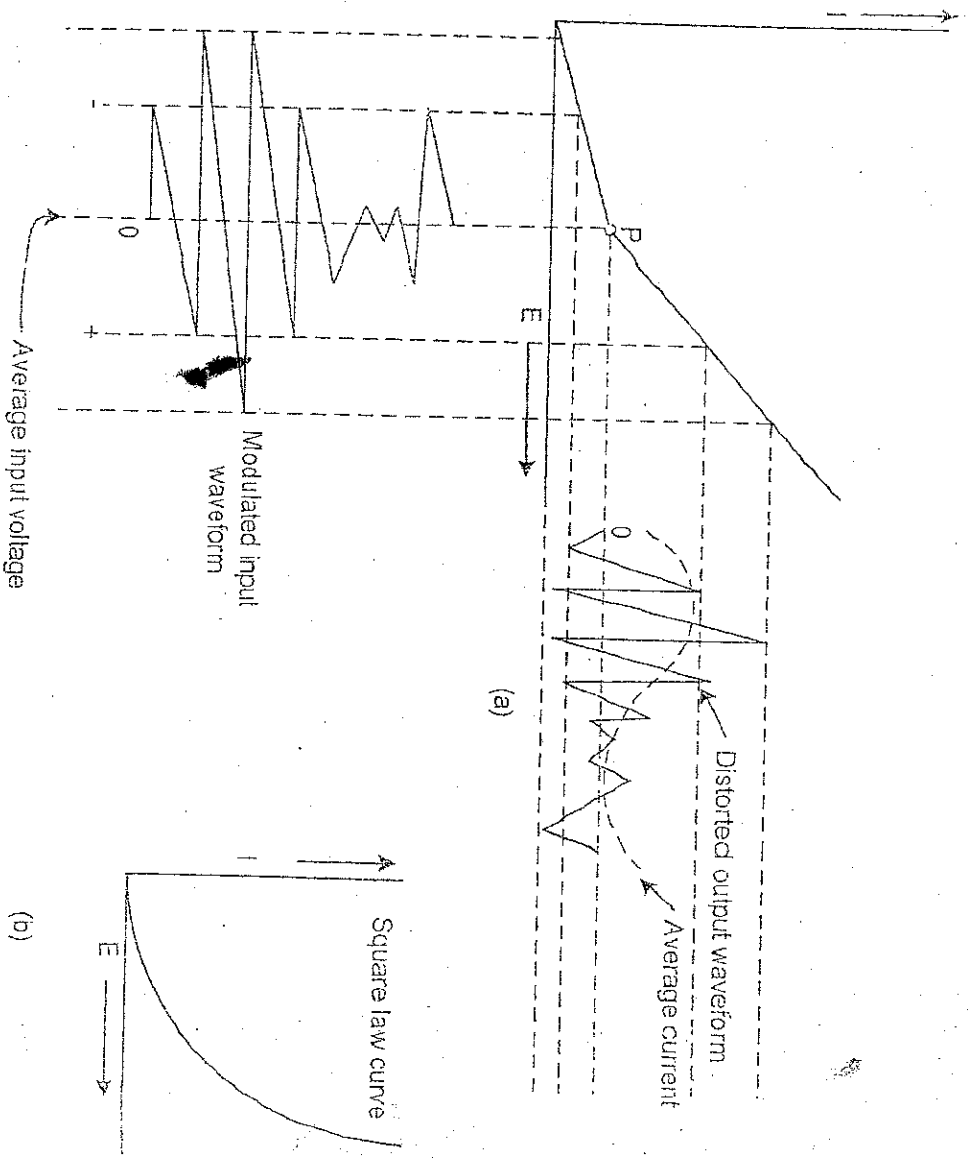


Figure 19: Nonlinear Device used as Detector — *Ideal Curve*

When the input to an ideal nonlinear device is a carrier and its sidebands, the output contains the following frequencies:

1. The carrier frequency
2. The upper sideband
3. The lower sideband
4. A dc component
5. A frequency equal to the carrier minus the lower sideband and the upper sideband minus the carrier, which is the original signal frequency.

The detector reproduces the signal frequency by producing a distortion of a desirable kind in its output. When the output of the detector is impressed upon a low-pass filter, the radio frequencies are suppressed and only the low-frequency intelligence signal and dc components are left. This is shown as the dashed average current curve in Figure 19(a). In some practical detector circuits, the nearest approach to the ideal curve is the square-law curve shown in Figure 19(b).

The Super-heterodyne Radio Receiver

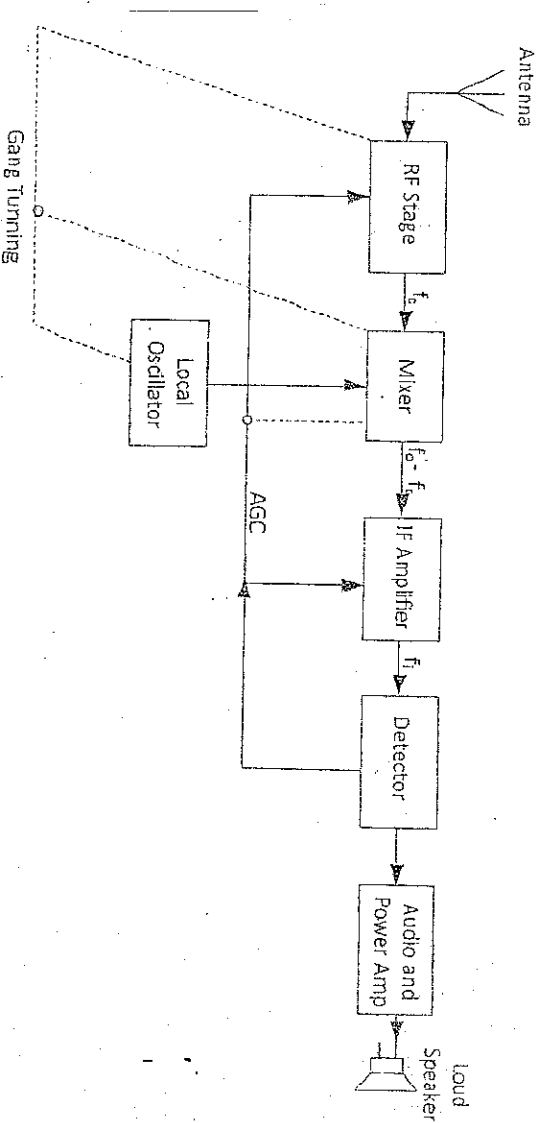


Figure 20: Superheterodyne Radio Receiver

The block diagram of a superheterodyne radio receiver is shown in figure above. In the superheterodyne receiver, the signal voltage is combined with the local oscillator voltage and normally converted into a signal of lower fixed frequency. This signal at the intermediate frequency contains the same modulation as the original carrier, and it is

amplified and detected to reproduce the original information. The superheterodyne thus has the same essential components as the TRF receiver in addition to the mixer, local oscillator and intermediate frequency (IF) amplifier. A dc level proportional to the received signal's strength is extracted from the detector stage and fed back to the IF amplifiers and sometimes to the mixer and/or the RF amplifier. This is the automatic gain control (AGC) level, which allows relatively constant receiver output for widely variable radio signals.

A constant frequency difference is maintained between the local oscillator and the RF circuits, normally through capacitance tuning in which all the capacitors are ganged together and operated in unison by one control knob. The IF amplifier provides most of the gain (sensitivity) and bandwidth requirements of the receiver; that is, it provides the bulk of radio-frequency signal amplification at a fixed frequency. This allows for constant bandwidth over the entire band of the receiver and is the key to superior selectivity of the superheterodyne receiver over the TRF receiver. Selectivity and sensitivity of the superhet are usually fairly uniform throughout its tuning range and is not subjected to the variations that beset the TRF receivers. The RF circuits are now used mainly to select wanted frequency, to reject interference such as image frequency, and to reduce the noise-figure of the receiver especially at high frequencies.

Reason for use and functions of RF amplifier: A receiver having an RF stage is undoubtedly superior in performance to the receiver without one, all being equal. On the other hand, there are some instances in which an RF amplifier is uneconomical, that is, where its inclusion would increase the cost of the receiver significantly while improving performance marginally.

The advantages of having RF amplifier are as follows:

1. Greater gain, that is, better sensitivity.
2. Improved image frequency rejection.
3. Improved signal-to-noise ratio.
4. Improved rejection of adjacent unwanted signals, that is, better selectivity.
5. Better coupling of the receiver to the antenna (important at VHF and above).

Image Frequency and its Rejection: The local oscillator frequency is usually made higher than the station or signal frequency, f_s , by an amount equivalent to the intermediate frequency (IF). That is, $f_o - f_s = f_i$. Then if another frequency f_{si} is picked by the receiver, the mixer circuit will together with the local oscillator frequency f_o produce a difference

$$\begin{aligned}
 f_p &= f_s + f_i \\
 f_{si} &= f_o + f_i \\
 f_{si} &= f_s + f_i + f_i = f_o + 2f_i
 \end{aligned}$$

frequency f_i' which resembles the original intermediate frequency f_i . The signal frequency f_{si} that causes the production of the frequency f_i' is called image frequency. Thus image frequency is the unwanted signal frequency which if allowed to enter the radio receiver would cause interference.

Image frequency rejection is the ability of the radio receiver to put off the unwanted signal frequency outside its operating band, so as not to interfere with the operating station frequency. The tune circuit of the mixer makes the rejection very possible because of its particular characteristics. A worked example will illustrate it better.

Example: Determine the image frequency for a standard broadcast band radio using a 455 kHz IF which is tuned to a 650 kHz station. Show whether image frequency rejection will be possible through your results and the tuned circuit characteristics.

Solution: Image frequency is given by:

$$f_{si} = f_s + 2f_i; \text{ But } f_s = 650 \text{ kHz (Station frequency):}$$

$$\text{For 455 kHz IF, } f_{si} = 650 + 2 \times 455 = 1560 \text{ kHz.}$$

$$\text{But the allowable band for station frequency of 650 kHz } = 2f_i = 2 \times 650 \text{ kHz} = 1300 \text{ kHz.}$$

Within the band, there will be interference. Outside the band, there will be no interference. Thus for the 455 kHz IF, image frequency rejection is possible since the image frequency is outside the operating band of the station. This is shown below by the mixer tuned circuit characteristics.

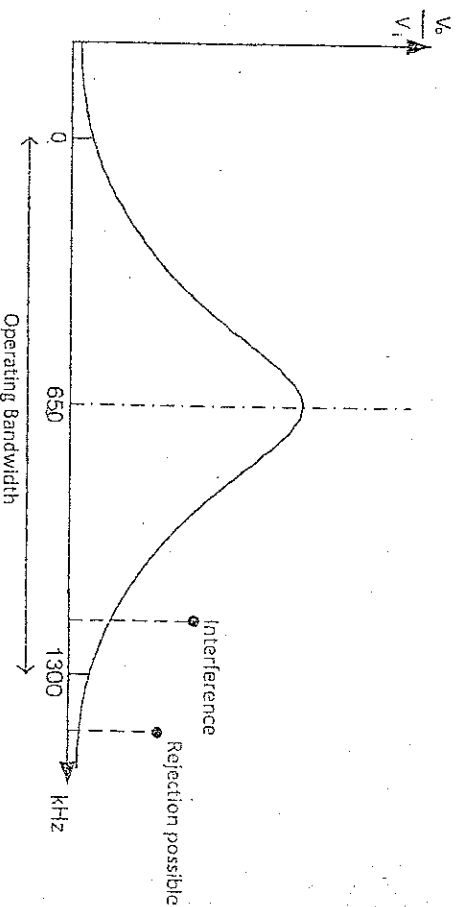


Figure 21: Mixer Tuned Circuit Characteristics

Exercise: ~~2019/20~~

Determine the image frequency for a standard broadcast band radio using 450 kHz IF with another using 250 kHz IF. Both are tuned to a 700 kHz station. Determine the station bandwidth. State whether image frequency rejection is possible in both cases. Use the mixer tuned circuit characteristics and your result to illustrate your point.

TELEVISION

Introduction: The word "Television" is derived from two words, "Tele" meaning 'distance' and "Vision" meaning 'seeing'. Hence Television actually means 'seeing at a distance'. The concept of television was developed in the 1920s. Commercial broadcasting started in the 1940s. In Nigeria it started in the late 1950s in the present South West zone, anchored by the late Chief Obafemi Awolowo.

Transmitter Principles

A television transmitter is made up of two separate transmitters – the sound transmitter and the video or picture transmitter. The sound transmitter is an FM system similar to broadcast FM radio; while the video or picture transmitter utilises the AM technique. During transmission the composite transmitted signal is a combination of both AM and FM principles. This is done to minimize interference effects between the two at the receiver because FM receiver is relatively insensitive to amplitude modulation and an AM receiver has rejection capabilities to frequency modulation.

The Simplified TV System

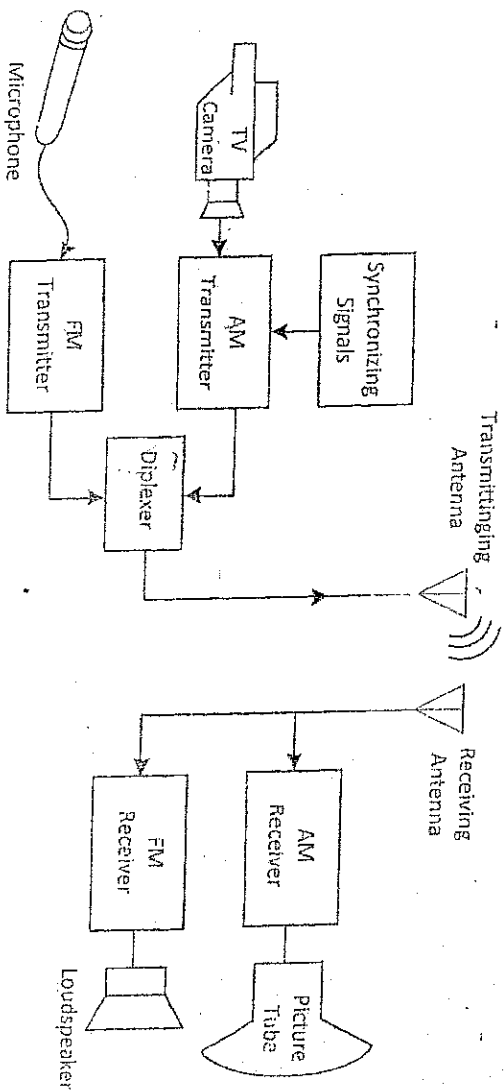


Figure 22: Simplified Television System

Figure 22 shows a simplified block diagram of a TV system. The TV camera converts the picture scene into an electrical signal. Thus the camera is thus a transducer that converts light energy into electrical energy. At the receiver, the CRT picture tube another transducer that converts the electrical energy back into light energy. The microphone and the loudspeaker are similarly related transducers for sound.

transmission and reception. The transmitting antenna converts the electrical energy into electromagnetic energy required for transmission through space, while the receiving antenna converts the electromagnetic energy into electrical energy for the receiver to process into light energy.

The diplexer is feeding the transmitting antenna with both video and sound signals while not allowing either to be fed back into one another's transmitter.

Scanning

The most widely used pickup device in TV cameras is the *charge couple device* (CCD). The CCD is a solid state chip consisting of thousands or millions of photosensitive cells arranged in a two-dimensional array. When light (photons) strike the CCD surface, the light information is converted to an electronic analog of the light.

Scanning is like reading a book. It starts from the upper left hand corner of the screen and moves to the right hand side of the screen tilting slightly downwards to the end of the line. The scene to be televised is focused onto the photosensitive cells in the CCD imaging device. For simplicity sake for example the letter "T" is picked. Instead of millions of cells, let it be assumed that the system has 30 cells arranged in 6 rows and 5 columns. Each separate cell is called a *pixel* meaning "picture element". The greater the number of pixels, the better the quality (or resolution) of the transmitted picture.

As the scene letter "T" is focused on the light-sensitive area so that all of rows 1 and 6 are illuminated, while all of row 2 is dark and the centers of rows 3, 4 and 5 are dark.

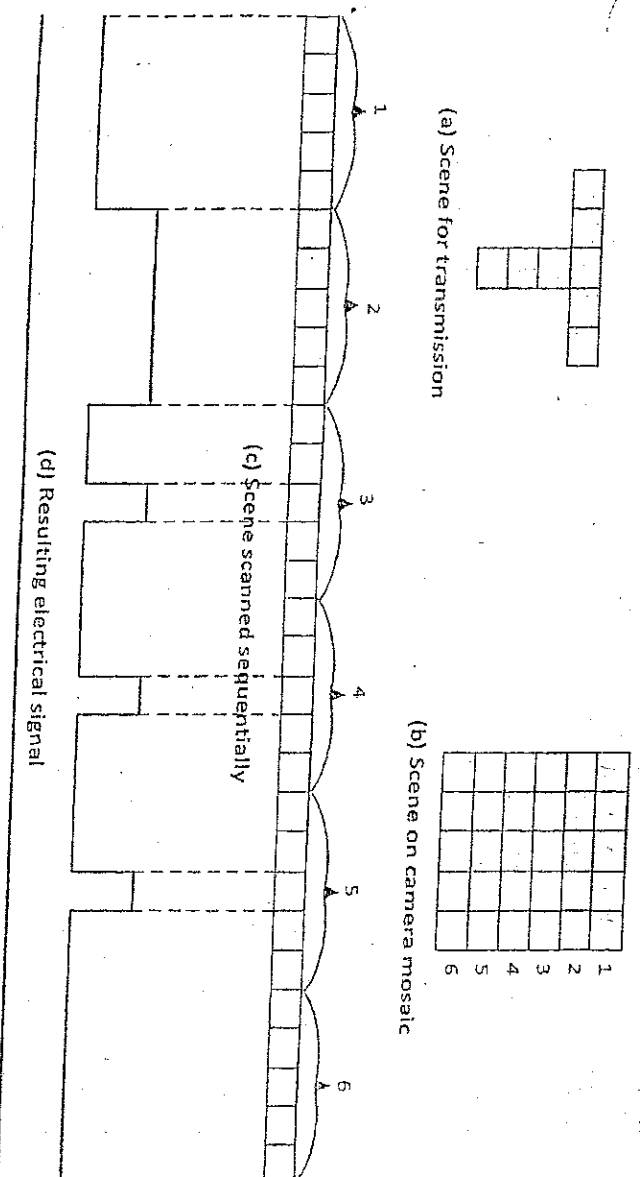


Figure 23: Simplified Scanning representation

Now if the rows of the scene are scanned sequentially and if the retrace interval is essentially zero, then Figure 23(c) shows the sequential breakup of information. The *Retrace Interval* is the time it takes an electron beam to move from the end of one line to the start of the next lower line. It is usually accomplished very rapidly. The variable light on the photosensitive cells results in a similar variable voltage being developed at the CCD's output, as shown in Figure 23(d).

Interlaced Scanning

The *frame frequency* is the number of times per second that a complete picture scene is traced. The rate for broadcast TV is 30 times per second. In other words, a complete scene (frame) is traced every $\frac{1}{30}$ s. Thirty frames per second is not enough to keep the human eye from perceiving *flicker* as a result of non-continuous visual presentation. If the frame frequency were increased to 60 frames per second, the flicker

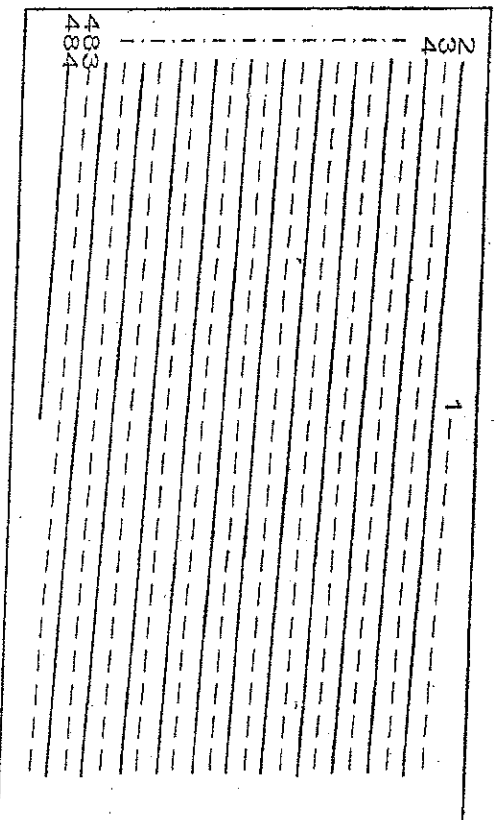


Figure 24: Interlaced Scanning

would no longer be apparent, but the video bandwidth would have to be doubled which would be very costly. Instead, the process of *interlaced scanning* is used to trick the human eye into believing it is seeing 60 pictures per second.

Figure 24 illustrates the process of interlaced scanning. The first set of lines (even field) is traced in $\frac{1}{60}$ s, and then the second set of lines (odd field) that make up the full scene (485 lines total) is interleaved between the first lines in the next $\frac{1}{60}$ s. Therefore lines 2, 4, 6, etc., occur during the even field, with lines 1, 3, 5, etc., interleaved between

the even numbered lines in the odd field. The field frequency is thus 60 Hz, with a frame frequency of 30 Hz.

Horizontal Synchronization

To accommodate the 525 scanning lines (485 visible) every $\frac{1}{30}$ s, the transmitter must send a synchronization (sync) pulse between every line of video signal so that perfect transmitter-receiver synchronization is maintained. The detail of these pulses is

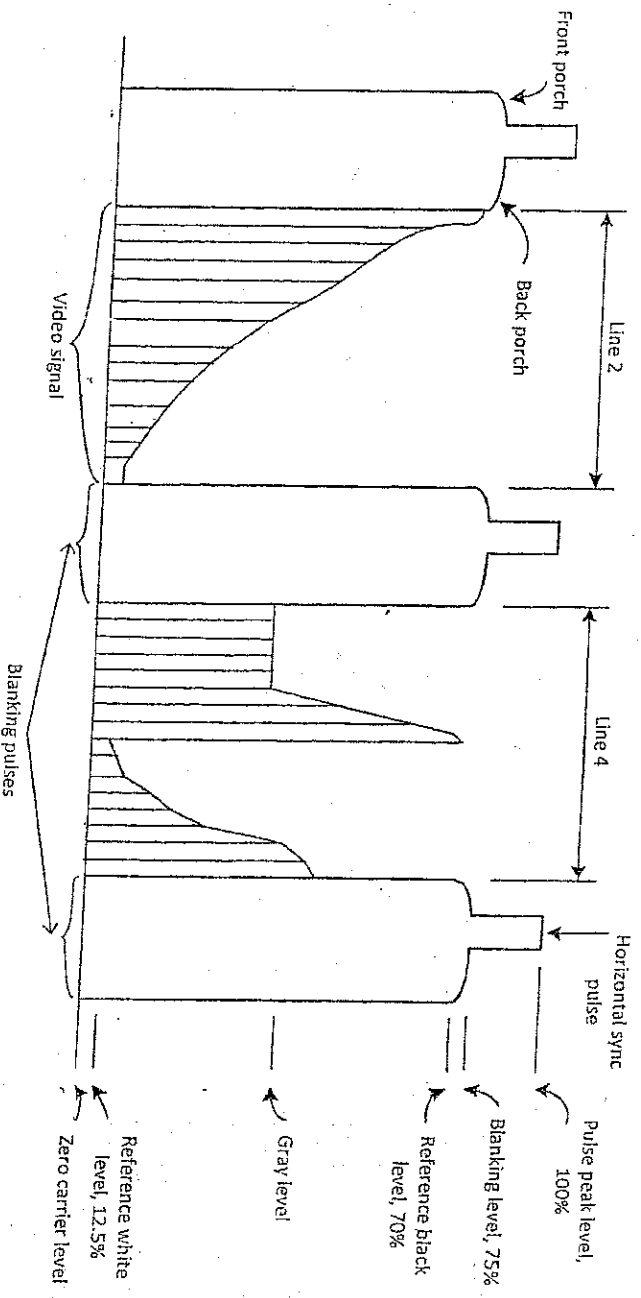


Figure 25: Horizontal blanking and sync pulses

shown in Figure 25. Three horizontal sync pulses are shown along with video signal for two lines. The sync pulses are superimposed on top of the horizontal blanking pulses as shown in the figure. The *blanking pulse* is a strong enough signal that is sent alongside the video signal during transmission so that the electron beam retrace at the receiver is “blackened” out and thus invisible to the viewer. The interval before the horizontal sync pulse appears on the blanking pulse is termed the *front porch*, while the interval after the end of the sync pulse, but before the end of the blanking pulse, is called the *back porch*.

The two lines of video picture signal shown in Figure 25, can be described as follows:
Line 2: It starts out nearly full black at the left-hand side and gradually lightens to full white at the right-hand side.

Line 4: It starts out medium gray and stays there until one-third of the way over, and it gradually becomes black at the picture center. It suddenly shifts to white and gradually turns darker gray at the right-hand side.

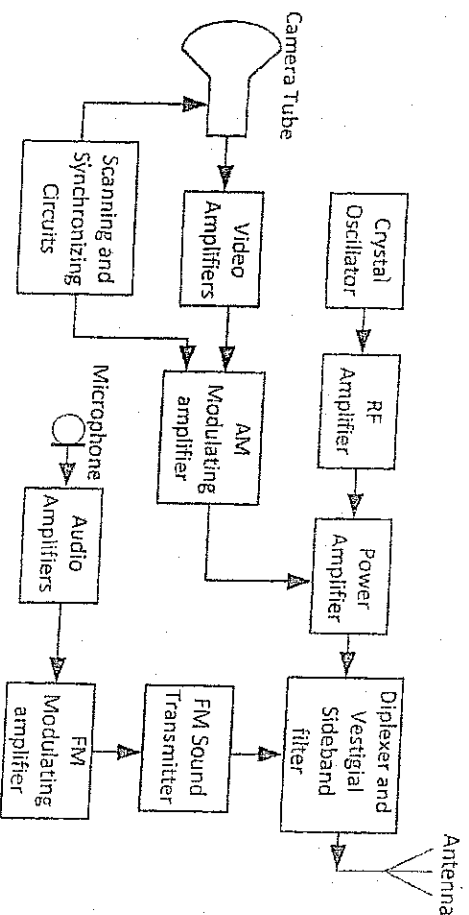


Figure 26: Black and white (Monochrome) TV transmitter

Colour Television

The introduction of colour television presents a much greater sophistication into the television engineering. One area of challenge was to make the black and white TV compatible with the colour TV, and this was successfully accomplished.

Compatibility of colour TV systems means that a colour TV system signal can be transmitted and be received by a black and white TV system. Also a black and white TV signal can be transmitted to be displayed in the colour TV system.

Conditions for compatibility: For compatibility to take place, the colour television system must:

1. Transmit, and be capable of receiving a luminance signal which is either identical to a monochrome transmission, or easily converted to it.
2. Use the same 6 – MHz bandwidth as monochrome TV.
3. Transmit the chroma information in such a way that it is sufficient for adequate colour reproduction, but easy to ignore by a monochrome receiver in such a way that no interference is caused by it.

Principles of Colour Television

At the colour TV transmitter the scene to be televised is actually scanned by three separate pickup tubes in the camera, each being sensitive only to one of the three

$Y = 1$ — Black & white (4 MHz)
 $I = 1.2$ } (3.8 MHz) Color TV
 $Q = 1.04$

PAL - phase Alternating Line.

NTSC - National Television System Committee

primary colours, red, blue and green. Since the various combinations of these three colours can be mixed to form any colour to which the human eye is sensitive, an electrical representation of a complete colour scene is possible. The three colour cameras scan the scene in unison, with the red, green and blue colour content separated into three different signals. At the receiver these three separate signals are made to illuminate groups of red, green and blue phosphor dots (called *triads*), and the original scene is reproduced in colour.

After generation these three separate colour signals are fed into the transmitter signal processing circuits (*matrix*) and create the Y, or *luminance* signal and *chroma*, or colour signals I and Q. The Y signal contains just the right proportion of red, blue and green such that it creates a normal black-and-white picture. It modulates the video carrier just as does the signal from a single black-and-white camera with a 4-MHz bandwidth. The chroma signals, I and Q, are used to phase modulate the 3.58-MHz colour subcarrier, which then *interleaves* their colour information in the gap left by the luminance Y signal's sidebands. This modulation by I and Q signals is accomplished by a balanced modulator, thus suppressing the 3.58-MHz subcarrier, as it would cause interference at the receiver.

At the receiver, a monochrome set will simply detect the Y signal and thus present a normal black-and-white rendition of a colour picture. The chroma signals (I and Q) cannot be detected in a monochrome set because their 3.58-MHz subcarrier has been suppressed and is not present in the received signal. Thus a colour TV set must have a means to generate and reinject the 3.58-MHz subcarrier to enable detection of the I and Q signals.

$$Y = 0.30R + 0.59G + 0.11B$$

$$I = 0.60R - 0.28G - 0.32B$$

$$Q = 0.21R - 0.52G + 0.31B$$

Additional Notes on TV System

Persistence of Vision: Persistence of vision is the ability for the human eye to retain light in a fraction of time, when the scene changes from light to darkness, or the ability for the eye to continue to see darkness in a moment when the scene changes from darkness to light.

Table: Standards of Major Television Systems

Standard	American System	European System
1. Number of lines per frame	525	625
2. Number of frames per second	30	25
3. Field frequency, Hz	60	50
4. Line frequency, Hz	15,750	15,625
5. Video bandwidth, MHz	4.2	5

Exercise: Using the above table solve this problem:

The line frequency of a standard TV is given as 18250 Hz. If the number of lines per frame of picture is 625, what is the error associated with the line frequency when the number of picture frames per second is 25 and why is it so? Which country is the TV from? Mention the name of its monochrome and colour TV systems. — *Pal*

The American colour TV system is called what? — *NTSC*

$$625 \times 25 = 15625$$

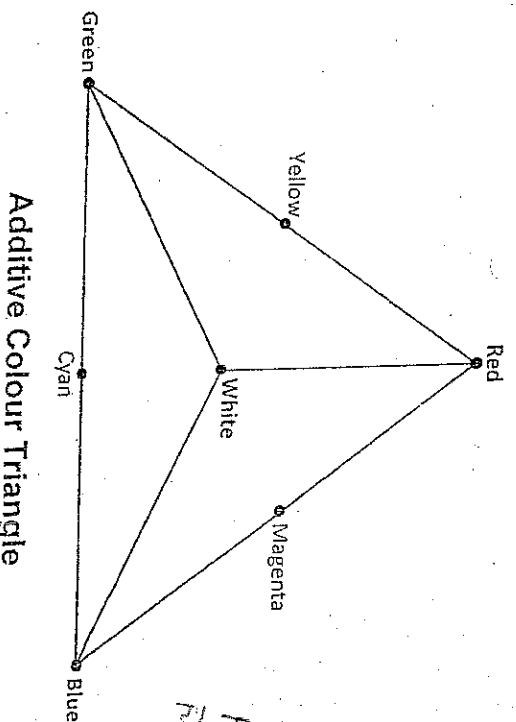
$$18250 - 15625 = 2625$$

Synchronization in TV systems: Synchronization in television is a process whereby the speed of the transmitter scanning must be exactly duplicated by the receiver scanning process in order to avoid distortion in the receiver output. For example, when the TV camera starts line 1, the receiver must also start line 1 on the CRT output display.

Aspect Ratio: This is the ratio of the width of the picture to the height of it, and it is standardized to 4:3. It is the most pleasing picture orientation to the human eye.

Colour Mixing in TV Engineering: The type of colour mixing used in TV engineering is called additive colour mixing, while that of pigments is called subtractive colour mixing.

The Colour Triangle:



*Radio & Television
For Test*

Mathematical Relationship between Colour signals and primary Colour Components:

$$Y = 0.30R + 0.59G + 0.11B$$

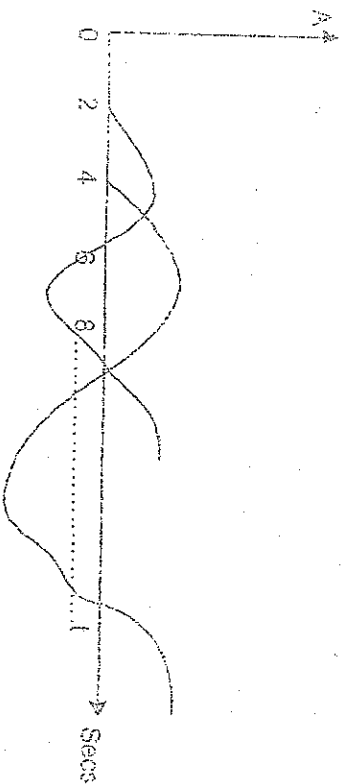
$$I = 0.60R - 0.28G - 0.32B$$

$$Q = 0.21R - 0.52G + 0.31B$$

Holly - B-5750

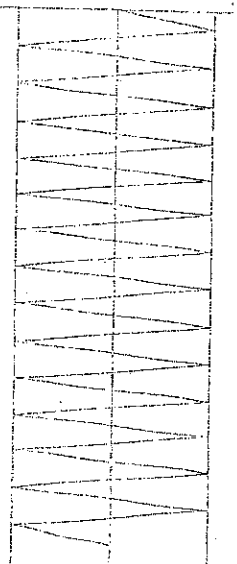
TEST 1 MODEL ANSWERS: IND II 2013/2014 SESSION DATE: 6/06/2016

(a)(i)

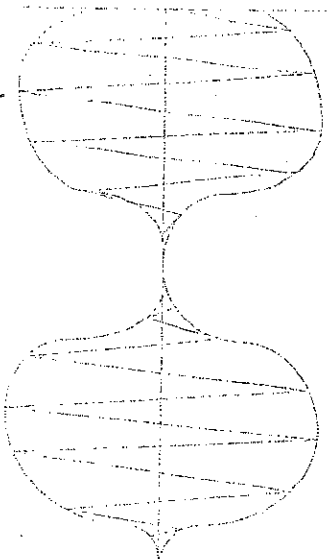


Given $f_1 = 5 \sin \omega t$, $f_2 = 5 \sin \omega(t+2)$ secs; $f_2 = 5 \sin \omega(t+4)$ secs

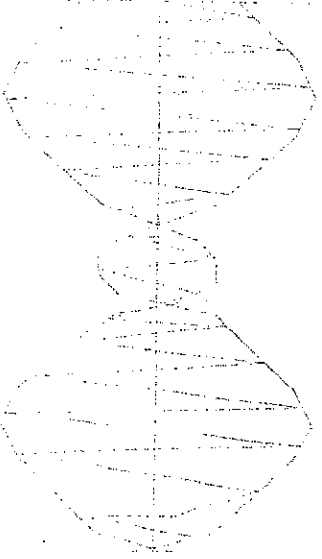
(ii)



0% Modulation



100% Modulation



120% Modulation

